AUTONOMOUS VEHICLES & DRONES

Research Report
Robotics & Automation Internship — Task 1
CodeAlpha

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ABSTRACT

Autonomous vehicles (AVs) and unmanned aerial vehicles (UAVs), commonly referred to as drones, represent two of the most transformative technological developments of the 21st century. Driven by advances in artificial intelligence, sensor fusion, and embedded computing, these systems are rapidly transitioning from research laboratories to real-world deployment across transportation, logistics, healthcare, agriculture, and defense sectors.

This report provides a comprehensive overview of the underlying technologies, key applications, real-world case studies, and critical challenges associated with AVs and drones. The findings indicate that while significant progress has been made with companies such as Waymo, Tesla, Amazon, and DJI demonstrating commercial viability widespread adoption remains constrained by regulatory, safety, and infrastructure barriers. The report concludes with forward-looking recommendations aligned with the trajectory of smart city development and next-generation mobility systems.

1. INTRODUCTION

The concept of a machine that can navigate the physical world without direct human input has captured the imagination of engineers, policymakers, and the public alike for decades. Today, this concept is no longer hypothetical. Autonomous vehicles operate on public roads in cities like San Francisco and Phoenix, while delivery drones complete thousands of commercial flights every week in Australia, Iceland, and the United States.

From the perspective of Robotics and Intelligent Systems, AVs and drones are among the most sophisticated real-world embodiments of autonomous agents. They integrate perception (sensing the environment), cognition (decision-making), and action (physical movement) — the three pillars of intelligent robotic behavior. Understanding these systems is therefore not merely an academic exercise; it is essential preparation for the next generation of robotics engineers. The concept of a machine that can navigate the physical world without direct human input has captured the imagination of engineers, policymakers, and the public alike for decades. Today, this concept is no longer hypothetical. Autonomous vehicles operate on public roads in cities like San Francisco and Phoenix, while delivery drones complete thousands of commercial flights every week in Australia, Iceland, and the United States.

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2. METHODOLOGY

This report is based on a systematic review of secondary research sources. The information presented was gathered through:

- Peer-reviewed and technical literature from IEEE Xplore, ScienceDirect, and ResearchGate
- Industry white papers and technical documentation from Waymo, Tesla AI, Amazon, and DJI
- Reports and regulatory frameworks published by the SAE International, FAA, EASA, and CAAC
- Technology news outlets including MIT Technology Review, IEEE Spectrum, and The Verge

3. CORE TECHNOLOGIES

AVs and drones both share a common technology stack, even though they operate in different environments.

3.1 Sensors

No single sensor gives a complete picture of the world. That is why autonomous systems use sensor fusion combining data from multiple sensors into a unified environmental model. The main ones are LiDAR (laser-based 3D mapping), radar (reliable in bad weather), cameras (visual detail and color), GPS with HD maps, and IMUs (measuring motion and orientation). Tesla controversially uses cameras only, arguing a sufficiently powerful neural network can match human vision. Waymo uses all of the above.

3.2 AI & Perception and Decision-Making

Convolutional Neural Networks (CNNs) process sensor data to detect and classify objects pedestrians, traffic signs, other vehicles. Transformer-based models are increasingly used for predicting how those objects will move next, which is critical for safe maneuvering in traffic. The planning stack then converts that understanding of the world into actual vehicle movements: route planning at the high level, then behavioral planning (when to merge or yield), then motion planning at the actuator level.

3.3 Simultaneous Localization and Mapping (SLAM)

Simultaneous Localization and Mapping (SLAM) let an autonomous agent build a map of its environment while figuring out where it is within that map at the same time. For drones operating where GPS does not work indoors, underground, dense urban areas. Visual-Inertial SLAM using cameras and IMU data is the main navigation method. AVs use LiDAR SLAM to match live sensor readings against pre built HD maps.

4. AUTONOMOUS VEHICLES

SAE International defines six levels of driving automation, which are now universally used to classify AV capability:

Level	Name	Description	Example
L0	No Automation	Human controls everything	Conventional cars
L1	Driver Assist	System helps with steering OR speed	Cruise control, ABS
L2	Partial Auto	System handles both; human stays alert	Tesla Autopilot, GM Super Cruise
L3	Conditional Auto	System drives; human takes over on request	Honda Sensing Elite (Japan)
L4	High Auto	Fully driverless in defined zones	Waymo One robotaxi (US cities)
L5	Full Auto	No human input, any condition	Not yet achieved commercially

Table 1: SAE J3016 Levels of Driving Automation (SAE International, 2021)

4.1 Robotaxis

Waymo is currently the clearest proof that Level 4 autonomy works at commercial scale. As of late 2025, it was completing 450,000 paid driverless rides per week across San Francisco, Los Angeles, Phoenix, Austin, and Atlanta, with plans to expand to 10+ more US cities and London in 2026. The company crossed 100 million fully autonomous miles in mid-2025 doubling that milestone in roughly six months, which gives a sense of how fast the pace of deployment is accelerating.

Tesla launched its own robotaxi service in Austin in June 2025 using a small fleet of Model Y vehicles. By January 2026, the service transitioned to genuinely unsupervised operation. Early rides uncovered real issues wrong-side driving, phantom braking, passengers dropped in intersections and seven incidents were reported to NHTSA through October 2025. Tesla's approach is camera-only, betting that fleet-scale training data will eventually outperform sensor-heavy systems like Waymo's.

4.2 Commercial & Freight

Autonomous trucking is moving quickly in the commercial space. Aurora Innovation began unmanned commercial freight operations on Texas highways in April 2025, and the economic logic is compelling: autonomous trucks can operate significantly longer hours than human drivers, with estimates suggesting 30–40% reductions in long-haul freight costs at full scale.

5. DRONES (UNMANNED AERIAL VEHICLES)

The global UAV market was valued at $41.27 billion in 2025 and is projected to reach $47.55 billion in 2026, growing at a 16.4% CAGR. Fully autonomous systems are the fastest-growing segment. Drone types range from small consumer quadcopters to large fixed-wing military platforms, but the most commercially relevant are multi-rotor delivery drones and hybrid VTOL aircraft.

5.1 Delivery

Amazon Prime Air received FAA BVLOS approval in 2022 and has been operating commercially in Texas and California. Wing Alphabet's drone delivery arm has been running regular deliveries in Australia, Finland, and the US, and is expanding into Walmart deliveries in 2026. Gartner projected over one million drones delivering retail goods globally by 2026, up from around 20,000 just a few years prior. Autonomous drone delivery potentially cuts last-mile parcel costs by 70%, which explains why every major logistics company is investing heavily here.

5.2 Agriculture

DJI's Agras T40 can carry 40 kg of liquid and spray up to 40 acres per hour with centimeter-level GPS precision. Multispectral imaging drones generate NDVI maps revealing crop health, water stress, and nutrient deficiencies across entire fields in minutes. Drone spraying reduces chemical usage by roughly 30% compared to traditional methods. China is leading this space the CAAC registered over 2 million drones by 2023 and agricultural demand continues to drive growth.

5.3 Emergency Response

Thermal imaging drones have proven effective in search and rescue operations where ground access is impossible the 2018 California Camp Fire being a well-documented example. Coastal rescue agencies in Australia, Spain, and the UK have deployed drones that drop inflatable life rafts to drowning swimmers, cutting response time from several minutes to under 60 seconds. In 2025, the FAA released a proposed framework (Part 108) for large-scale BVLOS drone operations, which will significantly expand what emergency services can do with drones.

6. CHALLENGES & LIMITATIONS

Challenge	**Key Issue**
Safety & Edge Cases	AI systems can fail on rare scenarios not in training data — a key cause of early AV incidents
Regulation	Laws vary widely by country; FAA Part 108 (BVLOS drones) was only proposed in 2025
Cybersecurity	Remote hijacking, GPS spoofing, and data privacy are active threats at scale
Public Trust	High-profile incidents (Tesla NHTSA reports, 2026) slow adoption despite overall safety gains
Infrastructure	Roads, cities, and airspace management systems were not designed for autonomous agents

Table 2: Key Challenges Facing Autonomous Vehicles and Drones

The edge case problem is probably the hardest one technically. These systems perform well on scenarios well-represented in their training data, but they can behave unpredictably in rare situations. The Tesla NHTSA reports in 2025–2026 show this is not just a theoretical concern real incidents happen when systems encounter things they were not adequately trained on. Waymo's approach is to run over 15 billion simulated miles per year alongside real-world testing specifically to expand coverage of rare scenarios.

7. ANALYSIS & DISCUSSION

Looking at AVs and drones together, the clearest pattern is that the technology has largely outpaced the ecosystem around it. The core technical capabilities perception, planning, control have been demonstrated convincingly in controlled environments. The harder problem now is not whether these systems can work, but how to integrate them into cities, airspace, and legal frameworks that were not designed with them in mind.

2026 is being called the 'year of autonomous' by several analysts, and that framing actually holds up when you look at the data. Waymo is operating in over 10 US cities with London on the way. Tesla entered commercial robotaxi service. Aurora launched unsupervised freight trucks on highways. Wing is scaling retail delivery with Walmart. These are not pilot programs anymore.

That said, there is a real tension between the speed of deployment and the depth of safety validation. Waymo has 100 million autonomous miles and a strong safety record. Tesla has a few thousand miles and a list of NHTSA reports. These are very different maturity levels being discussed as if they are the same thing. From an engineering standpoint, the challenge is that the public and regulators tend to judge the entire technology category based on the most visible incidents, regardless of which company caused them.

For drones, the regulatory environment is genuinely shifting. The FAA's proposed Part 108 rules in 2025 represent the first serious attempt to create a standardized framework for large-scale autonomous BVLOS operations in the US. Once those rules are finalized, expect the commercial drone market to accelerate significantly the main bottleneck right now is legal, not technical.

8. CONCLUSIONS

Autonomous vehicles and drones have crossed from research into commercial deployment, and the pace of that deployment is clearly accelerating. The core enabling technologies sensor fusion, deep learning perception, SLAM, and real-time motion planning are mature enough to support real-world operations at scale. Waymo's 450,000 weekly rides and Wing's retail delivery expansion are not projections; they are current operational facts.

The remaining challenges are real but solvable: regulatory frameworks are catching up, sensor costs are falling, and public trust while fragile tends to build as incident rates stay low over time. For anyone studying Robotics and Intelligent Systems, this field is probably the highest stakes real-world test of everything the discipline has to offer.

9. RECOMMENDATIONS

- Regulators should finalize the FAA Part 108 BVLOS framework and create equivalent international standards to unlock commercial drone deployment at scale.

- AV developers should prioritize explainability the ability to reconstruct and justify the system's decision-making after an incident to build the regulatory and public trust needed for continued expansion.

- Cities planning smart infrastructure upgrades should include UTM (Unmanned Traffic Management) systems in their design, treating drone corridors the same way road networks are planned.

- Academic programs in robotics should integrate AV/UAV simulation environments (like CARLA or AirSim) into coursework, given how central simulation has become to safety validation in industry.

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